

FOSSTON, MN, USA

WMO: 727505

Lat: 47.593N

Lon: 95.775W

Elev: 1277

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-18.4	-33.2	0.9	-23.6	-26.8	1.3	-18.6	27.3	17.5	24.9	15.1	4.1	180	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	87.6	71.1	83.6	68.3	81.3	67.1	74.0	82.6	71.6	80.5	69.4	78.3	12.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	122.7	80.6	68.4	109.5	77.8	66.0	100.6	75.5	38.4	82.7	36.2	80.3	34.4	78.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.0	18.9	DB	-33.0	91.3	8.3	3.8	-39.0	94.1	-43.9	96.3	-48.5	98.5	-54.6	101.3	(4)
(5)			WB	-32.9	78.1	7.6	3.0	-38.4	80.2	-42.8	82.0	-47.1	83.6	-52.7	85.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	7.1	10.7	25.1	40.7	53.7	63.7	67.8	65.9	58.1	43.9	27.4	13.1		
	DBStd	24.22	15.02	14.25	14.02	10.87	9.02	6.94	5.75	6.03	9.19	9.76	12.58	13.75		
	HDD50	5676	1331	1100	777	312	63	2	0	0	27	240	680	1144		
	HDD65	9487	1796	1520	1238	730	363	105	36	62	245	656	1127	1609		
	CDD50	1999	0	0	4	33	179	412	551	494	271	52	3	0		
	CDD65	334	0	0	0	1	14	65	122	90	39	3	0	0		
	CDH74	3230	0	0	0	13	189	607	1161	859	366	35	0	0		
	CDH80	811	0	0	0	2	49	135	306	216	96	7	0	0		
(14) Wind	WSAvg	7.9	8.4	8.1	8.2	8.7	8.7	7.1	6.1	6.2	7.6	8.7	8.8	8.5		
(15) Precipitation	PrecAvg	24.4	0.6	0.4	0.9	1.7	2.8	4.3	3.6	3.4	2.8	2.2	0.8	0.7		
	PrecMax	33.2	2.4	1.3	3.1	5.6	7.8	11.6	9.4	9.7	10.0	10.2	3.5	1.9		
	PrecMin	12.6	0.0	0.0	0.0	0.0	0.4	0.7	0.6	0.5	0.4	0.1	0.0	0.0		
	PrecStd	4.7	0.6	0.3	0.6	1.3	1.5	2.2	2.1	1.8	1.8	2.1	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	45.0	63.7	74.7	85.6	88.3	90.9	90.4	88.3	79.3	61.5	44.7		
		MCWB	35.1	40.3	53.3	54.7	63.6	71.3	73.3	72.4	69.4	61.2	50.4	40.2		
	2%	DB	35.7	38.7	54.3	68.5	79.4	83.7	86.3	84.4	82.1	70.4	54.2	37.3		
		MCWB	33.0	35.2	47.9	51.8	60.5	68.0	72.2	68.6	65.7	57.1	46.6	34.2		
	5%	DB	31.9	35.8	47.5	63.5	73.4	81.0	82.4	81.7	78.8	63.9	48.1	34.4		
		MCWB	30.1	33.0	42.3	49.5	57.2	65.8	69.9	67.3	64.2	53.4	42.2	32.2		
	10%	DB	27.8	31.6	42.9	58.6	70.1	77.3	81.0	79.2	73.2	58.9	43.4	30.4		
		MCWB	26.1	29.7	37.7	46.7	55.8	63.5	69.0	66.0	61.6	50.1	38.2	29.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	41.0	55.3	58.4	68.0	75.2	77.4	75.3	72.1	63.8	53.2	40.6		
		MCDB	39.7	44.2	60.2	70.0	78.2	83.6	86.5	84.9	82.8	72.3	58.9	43.7		
	2%	WB	32.7	35.8	47.8	53.6	63.9	71.6	74.8	72.3	68.8	58.9	47.2	35.3		
		MCDB	34.8	38.9	54.4	64.4	73.9	79.9	83.0	81.9	78.9	68.6	52.3	37.7		
	5%	WB	29.9	32.7	42.2	50.7	60.6	68.4	72.6	70.0	66.3	54.4	42.9	32.2		
		MCDB	31.4	34.9	47.0	61.5	70.3	76.9	80.8	79.1	74.8	62.3	48.2	34.2		
	10%	WB	26.3	29.6	38.1	47.7	57.9	66.1	70.3	67.9	63.9	51.0	38.9	29.7		
		MCDB	27.8	31.3	42.1	58.3	66.7	74.3	78.7	76.2	71.4	57.9	43.1	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.8	19.0	20.1	21.7	19.9	21.1	22.4	21.4	17.4	15.5	15.9		
		MCDBR	16.4	17.8	23.1	28.2	28.4	24.1	23.8	25.3	25.5	25.3	21.5	16.0		
	5% WB	MCWBR	14.3	14.8	16.8	15.9	14.8	12.7	13.7	13.3	13.3	14.6	14.6	13.5		
		MCWBR	16.1	16.1	22.0	25.3	22.9	20.3	21.9	21.8	21.9	22.9	20.5	14.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.279	0.319	0.295	0.350	0.367	0.377	0.380	0.381	0.351	0.305	0.278	0.267		
		taud	2.274	2.144	2.417	2.350	2.365	2.364	2.368	2.374	2.445	2.551	2.511	2.375		
	Ebn at Noon	Ebn at Noon	252	262	296	286	284	281	279	273	272	272	258	246		
		Edh at Noon	24	34	30	36	37	37	37	35	30	23	19	19		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	497	867	1224	1533	1708	1858	1978	1663	1221	748	446	350		
	RadStd	RadStd	52	81	127	138	106	134	91	123	89	115	47	40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-4.37	N/A	N/A	N/A	N/A	N/A	N/A	-214	-140
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page